ML Lab Week 14 - CNN Image Classification

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Introduction

Lab objective: given a dataset consisting of a total of 2188 images of hand gestures, categorised into 3 classes - rock, paper and scissors, process the images by resizing the images to 128x128, converting the images to PyTorch tensors and normalizing the pixel values, so that it ranges from -1 to +1 with mean and standard deviation 0.5, and build a CNN model. Train the model on these images and their corresponding labels. Validate the model's ability to classify the image correctly as 'rock', 'paper' or 'scissors'.

Model architecture

The CNN model is built using a convolutional block having 3 layers, each consisting of conv2d, relu and maxpool2d, and a fully connected block having 2 fully connected layers.
Convolutional block:

Layers →	1	2	3
conv2d Kernel: 3x3	Input: 3 channels, 128x128 Apply 16 filters Output: 16 feature maps	Input: 16 channels, 64x64 Apply 32 filters Output: 32 feature maps	Input: 32 channels, 32x32 Apply 64 filters Output: 64 feature maps
relu	Applies the activation function where negative pixel values are set to 0: relu(x) = max(x, 0)		
maxpool2d(2) Selects the largest pixel value out of a 2x2 sliding window with stride 2	Image is downsampled to 64x64	Image is downsampled to 32x32	Image is downsampled to 16x16

Fully connected block:
Receives the 3D 64x16x16 image to this block from convolutional block layer 3.

1. The 3d image is flattened into a $64 \times 16 \times 16 = 16384$ long 1d vector.
2. Layer 1 maps the 16884 features to 256 features and applies relu.
3. Dropout rate is set to 0.3, which means that this layer can randomly set 30% of the input features to 0.

4. The final fully connected layer maps the 256 features to 3 outputs - rock, paper, scissors.
5. The output with the highest output/score is the model's prediction for the given image.

Training and Performance

Key hyperparamters used:

Input image size: 128x128
Optimizer: Adam optimizer
Learning rate: 0.001
Loss function: Cross Entropy Loss
Epoch number: 10
Batch size: 32

The model was run on CUDA GPU
Final test accuracy: 98.86%

Conclusion and Analysis:

The Convolutional Neural Network (CNN) successfully learned to classify images of Rock, Paper, and Scissors. After training for 10 epochs, the model achieved a high test accuracy, demonstrating its effectiveness in distinguishing between the three hand gestures.

1. The training loss decreased consistently over 10 epochs (from 0.7871 to 0.0353), indicating that the model was learning and converging well on the training data. The relatively low final loss suggests the model was able to fit the training data effectively.
2. The model achieved an impressive 98.86% accuracy on the unseen test dataset. This high accuracy is a strong indicator that the model generalized well to new images it had not encountered during training, meaning it can reliably classify new Rock, Paper, Scissors images.
3. The predict_image function showed the model classified an image correctly as 'paper'.
4. The final code cell, simulating the game, also showed that the model correctly identified 'rock' and 'scissors' gestures.

Possible ways to improve:

- Data augmentation: rotations, flips, zooms, etc
- More CNN layers for better feature learning
- Experimenting with and tuning the hyperparamters
- L1/L2 regularization, early stopping to further reduce overfitting